

AI/ML Capstone Project Problem Statements

Parul University in collaboration with TelcoLearn

B.Tech 3rd/4th Year — 2027 Graduating Batch — Phase 2 Bootcamp

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Instructions for Students

Team composition: Each project is designed for a team of four students. Every sub-problem (Student A through D) is a self-contained contribution that one team member owns end-to-end, while the team integrates the parts into a single deployed application.

Full-stack requirement: Every project must be delivered as a running web application or mobile application. The deployment component is mandatory, not optional. Recommended stacks:

- **Web:** FastAPI or Flask (backend) + React or Streamlit (frontend), deployed on Render, Railway, or HuggingFace Spaces (free tier).
- **Mobile:** Flutter or React Native frontend calling a FastAPI backend; model served via ONNX Runtime or TFLite for on-device inference.
- **Student D (LLM integration):** Gemini API (Google AI Studio, free tier) or HuggingFace Inference API. Build the LLM component as a REST endpoint consumed by the frontend.

ML areas covered per project:

- **Student A** — Supervised learning (classification or regression with classical ML)
- **Student B** — Unsupervised learning (clustering, anomaly detection, or dimensionality reduction)
- **Student C** — Deep learning (CNN, LSTM, Transformer, or RL agent)
- **Student D** — LLM integration (prompt engineering, RAG, fine-tuning, or API orchestration) + full-stack deployment lead

Deliverables:

1. Kaggle/Colab notebooks for each sub-problem (reproducible, with outputs saved)
2. A deployed web or mobile application with a public URL or APK
3. A 10-minute demo video walking through the application and model results
4. A 4-page technical report (IEEE format) with model comparisons and architecture diagram

Evaluation criteria: Model performance (30%) · Code quality and reproducibility (20%) · Application design and usability (25%) · LLM integration quality (15%) · Presentation (10%)

Healthcare

Project 1 Cardiovascular Disease Risk Screening

Domain: Healthcare — *Dataset:* kaggle: sulianova/cardiovascular-disease-dataset

Problem statement: Build a web application that accepts a patient's clinical measurements (age, blood pressure, cholesterol, BMI, smoking status) and outputs a personalised cardiovascular disease risk score with an explanation of the top contributing factors. The application targets primary healthcare workers in rural clinics who lack specialist access.

Dataset: 70,000 patient records, 11 features, binary target (disease / no disease). Cardiovascular Disease Dataset, Kaggle ([sulianova/cardiovascular-disease-dataset](https://www.kaggle.com/sulianova/cardiovascular-disease-dataset)).

Student	Sub-problem	Technique	App component
A	Predict CVD risk (binary classification) with Logistic Regression, Random Forest, XGBoost. Tune decision threshold for high recall.	Supervised classification, GridSearchCV, AUC-ROC	Risk prediction API endpoint
B	Cluster patients into risk tiers using K-Means and Gaussian Mixture Models. Profile each tier's clinical characteristics.	Unsupervised clustering, silhouette score	Population risk dashboard
C	Build a PyTorch MLP with BatchNorm and Dropout. Compare Adam vs SGD convergence. Implement SHAP for feature attribution.	Deep learning, explainability	Per-patient SHAP waterfall chart
D	Prompt-engineer an LLM to translate the model's risk score and SHAP values into a plain-language clinical summary. Deploy full stack on Render (FastAPI + React).	LLM integration, full-stack deployment	Patient report generator, deployment

Project 2 Chest X-Ray Pneumonia Detection

Domain: Healthcare — *Dataset:* `kaggle: paultimothymooney/chest-xray-pneumonia`

Problem statement: Deploy a mobile application that allows a healthcare worker to photograph a chest X-ray and receive an instant pneumonia screening result with a Grad-CAM heatmap indicating which lung regions triggered the prediction. The system targets resource-constrained clinics where no radiologist is present.

Dataset: 5,863 chest X-ray images (Normal vs Pneumonia), JPEG format. Chest X-Ray Images, Kaggle ([paultimothymooney/chest-xray-pneumonia](https://www.kaggle.com/paultimothymooney/chest-xray-pneumonia)).

Student	Sub-problem	Technique	App component
A	Extract HOG and pixel histogram features; train SVM and Random Forest baseline. Establish non-deep-learning accuracy ceiling.	Supervised, feature engineering	Backend baseline comparison tab
B	Train CNN Autoencoder on normal X-rays only; flag high reconstruction error as anomalous (no disease labels used).	Unsupervised anomaly detection	Anomaly score overlay
C	Fine-tune DenseNet-121 on the full labelled dataset. Implement Grad-CAM heatmap visualisation on predicted images.	Transfer learning, XAI	Image upload + heatmap render
D	Build a radiology report generator: given prediction confidence and detected regions, an LLM produces a structured report. Convert model to TFLite; deploy as Flutter mobile app.	LLM, mobile deployment	Mobile camera interface, report PDF export

Project 3 Mental Health Risk Detection from Text

Domain: Healthcare — *Dataset:* kaggle: neelghoshal/reddit-mental-health-data

Problem statement: Build a web application that analyses social media posts or free-text journal entries, flags mental health risk signals, identifies emotional themes, and provides empathetic resources. Target users are campus counsellors and wellness coordinators who need to triage student mental health at scale.

Dataset: Reddit Mental Health Data, Kaggle (neelghoshal/reddit-mental-health-data); supplementary Mental Health Corpus (reihanenamdari/mental-health-corpus).

Student	Sub-problem	Technique	App component
A	Classify posts (depression / anxiety / neutral) using TF-IDF + Logistic Regression, Naive Bayes, SVM. Per-class F1 evaluation.	Supervised NLP classification	Text classification API
B	Apply LDA topic modelling to discover latent emotional themes. Visualise topic-word distributions using pyLDAvis.	Unsupervised topic modelling	Topic explorer tab
C	Fine-tune DistilBERT for 3-class mental health classification using HuggingFace Transformers. Compare against TF-IDF baseline.	Transformer fine-tuning	High-accuracy prediction service
D	Build a compassionate chatbot using Gemini API with a safety filter layer. Deploy as React web app with anonymous text input and a crisis resource panel.	LLM, safety guardrails, deployment	Chat interface, resource panel

Project 4 Drug Side Effect and Review Intelligence

Domain: Healthcare — *Dataset:* `kaggle: jessicali9530/kuc-hackathon-winter-2018`

Problem statement: Build a web platform that accepts a drug name, surfaces patient-reported side effects and sentiment patterns, clusters similar drugs by risk profile, and provides an LLM-generated plain-language interaction summary. Target users are patients and pharmacists who need to understand drug effects quickly.

Dataset: Drug Reviews Dataset, Kaggle (`jessicali9530/kuc-hackathon-winter-2018`), 215,000 patient reviews across 900+ drugs.

Student	Sub-problem	Technique	App component
A	Predict drug usefulness rating (regression) from review text using TF-IDF + Ridge Regression, XGBoost. RMSE and R^2 metrics.	Supervised regression	Rating predictor API
B	Cluster drugs by aggregated side effect word frequencies using K-Means and hierarchical clustering. Build a dendrogram visualisation.	Unsupervised clustering	Drug similarity explorer
C	Train LSTM and DistilBERT for review sentiment classification. Compare sequence model vs transformer on the same data.	Deep learning NLP	Sentiment analysis service
D	Build an LLM drug interaction checker: user inputs two drug names; system retrieves review summaries and the LLM explains potential combined risks. Deploy as Streamlit app.	LLM, retrieval, deployment	Drug checker interface

Finance

Project 5 Real-Time Credit Card Fraud Detection

Domain: Finance — *Dataset:* kaggle: `mlg-ulb/creditcardfraud`

Problem statement: Build a real-time fraud detection web application that processes transaction data, flags suspicious activity with an explanation, and provides a case management dashboard for fraud analysts. The system must handle extreme class imbalance (0.17% fraud rate) while minimising false positives.

Dataset: 284,807 transactions, 30 PCA-transformed features, 0.17% fraud rate. Credit Card Fraud Detection, Kaggle (`mlg-ulb/creditcardfraud`).

Student	Sub-problem	Technique	App component
A	Handle class imbalance with SMOTE and class weights. Train Logistic Regression, XGBoost. Optimise threshold for F1 on minority class.	Supervised, imbalanced learning	Transaction scoring API
B	Apply Isolation Forest and One-Class SVM for unsupervised anomaly detection without labels. Evaluate against ground truth.	Unsupervised anomaly detection	Anomaly score dashboard
C	Build a deep Autoencoder trained on legitimate transactions only; flag high reconstruction error as fraud. Tune threshold on validation set.	Deep learning autoencoder	Reconstruction error chart
D	Build a fraud analyst assistant: given flagged transaction features, an LLM generates a natural-language investigation memo. Deploy as FastAPI + React with a live transaction stream simulation.	LLM, full-stack deployment	Analyst memo generator, case queue

Project 6 Stock Market Signal Intelligence Platform

Domain: Finance — *Dataset:* `kaggle: camnugent/sandp500 + yfinance`

Problem statement: Build a financial intelligence web application that ingests stock price data, computes ML-based directional signals, identifies market regimes, forecasts short-term price movements, and integrates news sentiment for a holistic trading dashboard. Target users are individual investors and quant trainees.

Dataset: S&P 500 historical prices, Kaggle (`camnugent/sandp500`); live data via `yfinance` library; NIFTY 50 (`kaggle: rohanrao/nifty50-stock-market-data`).

Student	Sub-problem	Technique	App component
A	Engineer RSI, MACD, Bollinger Band features. Train XGBoost for next-day direction prediction using walk-forward validation.	Supervised time-series classification	Signal indicator panel
B	Apply PCA on 50+ technical features; cluster trading days into market regimes (bull/bear/sideways) using K-Means.	Unsupervised, dimensionality reduction	Regime indicator widget
C	Train a 2-layer BiLSTM for 5-day price forecasting. Report Directional Accuracy and RMSE vs ARIMA baseline.	Deep learning, sequence modelling	Price forecast chart
D	Scrape daily financial headlines; use Gemini API for sentiment scoring; combine signal with technical indicators into a composite score. Deploy as React dashboard with Recharts.	LLM, sentiment, full-stack	News sentiment feed, composite score

Project 7 Loan Default Prediction and Credit Scorecard

Domain: Finance — *Dataset:* `kaggle: home-credit-default-risk`

Problem statement: Build a web application for a microfinance institution that takes a loan applicant's financial and demographic information, outputs a credit score with default probability, and generates a structured credit memo explaining the decision. The system must be interpretable for non-technical loan officers.

Dataset: Home Credit Default Risk, Kaggle (`home-credit-default-risk`), 307,511 applications, 122 features across 7 relational tables.

Student	Sub-problem	Technique	App component
A	Feature engineer debt-to-income ratio, payment history aggregates. Train Logistic Regression and XGBoost. Produce calibrated probability scorecard.	Supervised credit scoring, calibration	Credit score API
B	Segment applicants by financial behaviour using K-Means and Gaussian Mixture Models. Profile high-risk cluster.	Unsupervised segmentation	Applicant segment tag
C	Build a deep tabular model with entity embeddings for categorical features in PyTorch. Compare Gini coefficient against XGBoost.	Deep learning tabular	Model comparison panel
D	Prompt an LLM to generate a structured credit memo from SHAP values and applicant summary. Deploy as React form + FastAPI backend on Railway.	LLM, explainability, deployment	Credit memo generator, PDF download

Project 8 Insurance Risk Assessment and Customer Advisory

Domain: Finance — *Dataset:* `kaggle: mirichoi0218/insurance`

Problem statement: Build a web application for insurance agents that accepts a customer's profile, predicts their risk tier and expected annual premium, explains the key premium drivers, and generates a personalised policy recommendation. The system replaces manual actuarial tables with a data-driven, explainable model.

Dataset: Medical Cost Personal Dataset (`kaggle: mirichoi0218/insurance`); Porto Seguro Safe Driver Prediction (`kaggle: porto-seguro-safe-driver-prediction`).

Student	Sub-problem	Technique	App component
A	Predict annual insurance premium (regression) using Ridge, Lasso, XGBoost. Log-transform target. Feature importance analysis.	Supervised regression	Premium prediction API
B	Cluster policyholders into risk tiers using K-Means. Map each tier to a renewal and upsell strategy.	Unsupervised segmentation	Customer tier widget
C	Train an MLP on tabular features. Implement SHAP waterfall plot for individual premium explanation.	Deep learning, XAI	SHAP waterfall visualisation
D	Build an LLM chatbot that explains a premium quote in plain language using SHAP values as context. Deploy as Streamlit app with a PDF quote generator.	LLM, deployment	Chatbot, quote PDF

Manufacturing and Industry

Project 9 Predictive Maintenance for Industrial Equipment

Domain: Manufacturing — *Dataset:* kaggle: stephanmatzka/predictive-maintenance-dataset

Problem statement: Build a web-based maintenance dashboard for a factory floor that monitors real-time machine sensor readings, predicts failure probability, identifies anomalous operating conditions, and generates a maintenance work order when failure risk exceeds a threshold.

Dataset: AI4I 2020 Predictive Maintenance Dataset, Kaggle (stephanmatzka/predictive-maintenance-dataset-ai4i-2020), 10,000 records, 5 failure types.

Student	Sub-problem	Technique	App component
A	Predict failure type (multiclass) using Random Forest and XGBoost. Optimise recall — a missed failure is more costly than a false alarm.	Supervised multiclass classification	Failure type alert panel
B	Apply DBSCAN on sensor readings to detect abnormal operating clusters without labels. Evaluate against known failure timestamps.	Unsupervised anomaly detection	Anomaly scatter plot
C	Train an LSTM on 24-hour windowed sensor sequences to predict failure 24 hours ahead. Early stopping and LR scheduling.	Deep learning, LSTM	24-hour forecast chart
D	Build a maintenance assistant: given sensor data, failure prediction, and past maintenance logs, an LLM generates a step-by-step work order. Deploy as FastAPI + React dashboard.	LLM, full-stack deployment	Work order generator, sensor live feed

Project 10 Visual Quality Control for Manufacturing Defects

Domain: Manufacturing — *Dataset:* `kaggle: ipythonx/mvtec-ad`

Problem statement: Build a mobile application for factory quality inspectors that accepts a product photograph from a phone camera, classifies the defect type, highlights the defect location on the image, and logs the defect to a central quality management dashboard. The system must run inference within 200ms on a mid-range Android device.

Dataset: MVTec Anomaly Detection Dataset, Kaggle (`ipythonx/mvtec-ad`); supplementary Steel Defect Detection (`kaggle: severstal-steel-defect-detection`).

Student	Sub-problem	Technique	App component
A	Extract HOG and LBP texture features. Train SVM and Random Forest for defective vs non-defective classification. Establish CV baseline.	Supervised image classification	Baseline comparison tab
B	Train CNN Autoencoder on defect-free images; flag high reconstruction error as defective. No defect labels needed.	Unsupervised deep anomaly detection	Anomaly score overlay
C	Fine-tune EfficientNet-B0 for defect classification. Implement Grad-CAM to localise the defect. Convert to TFLite for mobile inference.	Transfer learning, XAI, edge deployment	Mobile camera + heatmap view
D	Generate a quality report using an LLM given defect type, location, and production run data. Deploy Flutter app + FastAPI backend; log defects to SQLite quality dashboard.	LLM, mobile deployment	QC report, defect log dashboard

Project 11 Energy Consumption Forecasting for Smart Buildings

Domain: Manufacturing — *Dataset:* kaggle: uciml/electric-power-consumption-data-set

Problem statement: Build a web dashboard for building energy managers that displays real-time consumption, forecasts the next 24–48 hours of demand, identifies consumption anomalies, and provides LLM-generated energy-saving recommendations personalised to the building’s usage profile.

Dataset: Household Electric Power Consumption, Kaggle (uciml/electric-power-consumption-data-set), 2 million readings over 4 years at 1-minute resolution.

Student	Sub-problem	Technique	App component
A	Engineer lag and rolling statistics features. Train XGBoost for next-24-hour demand forecasting. Walk-forward validation. Report RMSE and MAPE.	Supervised time-series regression	24-hour forecast chart
B	Cluster households by daily load profile using K-Means on hourly averages. Identify demand response candidates.	Unsupervised load clustering	Profile cluster widget
C	Train a Seq2Seq LSTM for 48-hour multi-step forecasting. Compare with ARIMA and Prophet baselines.	Deep learning, sequence-to-sequence	48-hour multi-step forecast
D	Build an energy advisor using Gemini API: given a consumption profile and tariff schedule, generate personalised saving recommendations. Deploy as React dashboard with Chart.js.	LLM, full-stack deployment	Advisory panel, tariff comparison

Project 12 Supply Chain Demand Forecasting Platform

Domain: Manufacturing — *Dataset:* kaggle: rossmann-store-sales

Problem statement: Build a web-based supply chain intelligence platform for a retail chain that forecasts per-store daily demand, clusters stores by sales behaviour, generates automated purchase order recommendations, and provides a natural-language weekly summary for procurement managers.

Dataset: Rossmann Store Sales, Kaggle ([rossmann-store-sales](#)), 1.1 million rows of store-level daily sales with promotions and holidays.

Student	Sub-problem	Technique	App component
A	Feature engineer store, date, and promotion variables. Train XGBoost for store-level sales prediction using time-series CV.	Supervised demand forecasting	Per-store forecast panel
B	Cluster stores by weekly sales pattern using hierarchical clustering. Show dendrogram and cluster profiles.	Unsupervised time-series clustering	Store cluster map
C	Build an LSTM for per-store forecasting with store type and promotion as additional features. Compare against Prophet baseline.	Deep learning, LSTM	LSTM vs Prophet comparison chart
D	Build a procurement assistant: given forecast data and current stock, an LLM generates a natural-language purchase order. Deploy as Streamlit web app with CSV upload.	LLM, deployment	PO recommendation, inventory table

Agriculture and Environment

Project 13 Crop Disease Detection Mobile Application

Domain: Agriculture — *Dataset:* `kaggle: emmarex/plantdisease`

Problem statement: Build a mobile application for farmers that identifies crop diseases from a leaf photograph, provides treatment advice in local language, and logs disease outbreaks to a geospatial dashboard for agricultural officials to monitor disease spread across a region.

Dataset: PlantVillage Dataset, Kaggle (`emmaex/plantdisease`), 54,000 images, 38 disease classes across 14 crop species.

Student	Sub-problem	Technique	App component
A	Extract colour histogram and LBP texture features. Train SVM and Random Forest for disease classification. Per-class accuracy evaluation.	Supervised image classification	Baseline accuracy report
B	Apply PCA and t-SNE to visualise disease class separation in feature space. Cluster visually similar disease groups.	Unsupervised, dimensionality reduction	Disease similarity visualiser
C	Fine-tune MobileNetV2 for 38-class classification. Convert to TFLite (target: <200ms inference on Android). Report accuracy vs model size tradeoff.	Transfer learning, edge deployment	On-device image classifier
D	Given detected disease name and crop type, call Gemini API to generate treatment advice in Hindi or English. Deploy as Flutter mobile app + FastAPI backend with a Leaflet.js outbreak map.	LLM, multilingual, mobile deployment	Disease advice chatbot, outbreak map

Project 14 Air Quality Index Forecasting and Alert System

Domain: Environment — *Dataset:* kaggle: rohanrao/air-quality-data-in-india

Problem statement: Build a public-facing web application that displays city-level AQI forecasts for the next 48 hours, identifies high-pollution clusters, and automatically generates health advisory alerts tailored to different demographic groups (elderly, children, commuters).

Dataset: Air Quality Data in India, Kaggle (rohanrao/air-quality-data-in-india); UCI Air Quality Dataset (kaggle: fedesoriano/air-quality-data-set).

Student	Sub-problem	Technique	App component
A	Predict AQI category (Good / Moderate / Poor / Severe) using XGBoost. Engineer time-of-day and weather interaction features.	Supervised multiclass classification	AQI category forecast widget
B	Cluster cities by pollution profile using K-Means. Identify consistently high-pollution clusters and their primary pollutants.	Unsupervised clustering	City cluster pollution map
C	Train a multivariate LSTM on hourly readings (PM2.5, NO ₂ , CO, temperature, humidity) to forecast AQI for next 48 hours.	Deep learning, multivariate LSTM	48-hour AQI forecast chart
D	Given tomorrow's AQI forecast, use an LLM to generate tailored health advisories per demographic group. Deploy as Next.js web app with a city selector and push notification simulation.	LLM, full-stack deployment	Advisory panel, city selector

Mobility and Transportation

Project 15 Ride-Hailing Demand Forecasting and Driver Allocation

Domain: Mobility — *Dataset:* `kaggle: nyc-taxi-trip-duration + uber-pickups-in-new-york-city`

Problem statement: Build a web dashboard for a ride-hailing operations team that forecasts per-zone demand for the next three hours, displays a live driver allocation heatmap, and generates personalised driving recommendations for individual drivers seeking to maximise earnings.

Dataset: NYC Taxi Trip Duration, Kaggle (`nyc-taxi-trip-duration`); Uber Pickups NYC (`kaggle: fivethirtyeight/uber-pickups-in-new-york-city`).

Student	Sub-problem	Technique	App component
A	Predict trip duration (regression) using pickup/dropoff coordinates, time features, and weather. XGBoost with feature importance analysis.	Supervised regression	Trip duration predictor
B	Cluster pickup locations into demand zones using DBSCAN on geographic coordinates. Identify time-of-day zone demand patterns.	Unsupervised geo-clustering	Demand zone heatmap
C	Train an LSTM on hourly per-zone demand sequences to forecast next 3-hour demand. Feed into a driver allocation optimiser.	Deep learning, spatiotemporal LSTM	3-hour demand forecast
D	Build an LLM-based driver coaching assistant: given a driver's weekly trip history and zone patterns, generate personalised earning maximisation tips. Deploy as React dashboard with Mapbox GL.	LLM, geospatial, full-stack	Driver tips panel, interactive map

Project 16 Road Accident Severity Prediction and Hotspot Analysis

Domain: Mobility — *Dataset:* kaggle: tsiaras/uk-car-accidents-2005-to-2007

Problem statement: Build a web application for urban traffic authorities that predicts accident severity for a given road segment and time-weather condition, identifies accident hotspots on an interactive map, and generates infrastructure improvement recommendations for municipal engineers.

Dataset: UK Road Safety Accidents, Kaggle (tsiaras/uk-car-accidents-2005-to-2007), 1.5 million accidents over 15 years.

Student	Sub-problem	Technique	App component
A	Predict accident severity (Slight / Serious / Fatal) using Random Forest and XGBoost. Handle class imbalance with SMOTE.	Supervised multiclass classification	Severity predictor form
B	Identify geographic accident hotspots using DBSCAN on lat/long coordinates. Profile each cluster by dominant cause.	Unsupervised hotspot analysis	Hotspot cluster map
C	Build a Graph Neural Network (PyTorch Geometric) with road intersections as nodes. Predict accident probability per edge.	Deep learning, GNN	Graph-based risk overlay
D	Given hotspot cluster data and accident patterns, an LLM generates a structured infrastructure improvement report. Deploy as FastAPI + Leaflet.js web app with hotspot filtering.	LLM, geospatial, full-stack	Infrastructure report generator, map

Retail and E-Commerce

Project 17 Customer Churn Prediction and Retention Platform

Domain: Retail — *Dataset:* kaggle: blastchar/telco-customer-churn

Problem statement: Build a CRM web application for a telecom company that scores each customer's churn probability daily, segments the customer base, identifies the top churn drivers for each at-risk customer, and auto-generates a personalised retention offer via email.

Dataset: Telco Customer Churn, Kaggle (blastchar/telco-customer-churn), 7,043 customers, genuinely raw (TotalCharges dtype bug to fix).

Student	Sub-problem	Technique	App component
A	Fix raw dtype bug, encode categoricals, train Logistic Regression and XGBoost. Tune threshold for maximum recall on churners.	Supervised binary classification	Churn score API
B	Segment customers by usage and billing behaviour using K-Means. Identify at-risk segment profile.	Unsupervised segmentation	Segment distribution chart
C	Train an MLP with class weighting in PyTorch. Implement SHAP for top 10 churn drivers per customer.	Deep learning, XAI	SHAP waterfall per customer
D	Given a churning customer's profile and SHAP drivers, an LLM generates a personalised retention email and call script. Deploy as React CRM with SQLite customer database.	LLM, full-stack deployment	Retention email generator, CRM table

Project 18 Intelligent Product Recommendation System

Domain: Retail — *Dataset:* `kaggle: snap/amazon-fine-food-reviews + grouplens movielens-25m`

Problem statement: Build an e-commerce web application with a natural-language shopping assistant that understands what a user wants in free text, retrieves personalised product recommendations using a hybrid (collaborative + content-based) engine, and explains why each item was recommended.

Dataset: Amazon Fine Food Reviews, Kaggle (`snap/amazon-fine-food-reviews`), 568,000 reviews; MovieLens 25M, grouplens.org/datasets/movielens/25m.

Student	Sub-problem	Technique	App component
A	Build collaborative filtering using SVD matrix factorisation via the surprise library. Evaluate RMSE and NDCG@10. Address cold-start problem.	Supervised recommendation	Collaborative filter API
B	Content-based filtering using TF-IDF on product descriptions and cosine similarity clustering. Compare with collaborative filtering.	Unsupervised content filtering	Content-based recommender
C	Train Neural Collaborative Filtering (NCF) in PyTorch with user and item embeddings. Compare with SVD baseline on NDCG@10.	Deep learning, neural recommender	NCF recommendation API
D	Build an LLM shopping assistant: user types natural language query; LLM translates to filters; recommender returns results; LLM explains recommendations. Deploy as React storefront with product cards.	LLM, full-stack deployment	Chat interface, product grid

Project 19 Customer Review Sentiment Intelligence Dashboard

Domain: Retail — *Dataset:* kaggle: kazanova/sentiment140 + snap/amazon-fine-food-reviews

Problem statement: Build a brand monitoring web application that ingests customer reviews from multiple sources, classifies sentiment, discovers emerging complaint topics, tracks sentiment trends over time, and generates an automated weekly brand health report for product managers.

Dataset: Sentiment140, Kaggle (kazanova/sentiment140), 1.6 million tweets; Amazon Fine Food Reviews (snap/amazon-fine-food-reviews).

Student	Sub-problem	Technique	App component
A	TF-IDF + Logistic Regression, Naive Bayes, SVM for 3-class sentiment classification. Identify top predictive words per class.	Supervised NLP classification	Sentiment scorer API
B	LDA topic modelling on negative reviews to discover complaint themes. Visualise with pyLDAvis.	Unsupervised topic modelling	Topic explorer chart
C	Fine-tune DistilBERT for sentiment classification. Show accuracy gain over TF-IDF baseline. Analyse misclassifications.	Transformer fine-tuning	High-accuracy sentiment service
D	Build an LLM-powered weekly brand report generator: aggregate topic and sentiment data, generate an executive summary. Deploy as React dashboard with date-range filtering and PDF export.	LLM, full-stack deployment	Report generator, trend line charts

Education

Project 20 Student Performance Prediction and Early Intervention System

Domain: Education — *Dataset:* kaggle: larsen0966/student-performance-data-set

Problem statement: Build a web application for academic coordinators that monitors student engagement metrics, predicts dropout risk by week four of a semester, clusters students by behavioural pattern, and generates personalised study plans and motivational messages for at-risk students.

Dataset: Student Performance Dataset (UCI), Kaggle (larsen0966/student-performance-data-set); Open University Learning Analytics (analyse.kmi.open.ac.uk/open-dataset).

Student	Sub-problem	Technique	App component
A	Predict final grade and dropout risk using Logistic Regression, Random Forest, XGBoost on attendance and assignment scores.	Supervised classification + regression	At-risk flag API
B	Cluster students by learning behaviour (consistent, procrastinator, last-minute) using K-Means on submission timestamps and engagement.	Unsupervised behavioural clustering	Student segment dashboard
C	Train an LSTM on weekly engagement time series (logins, submissions, forum activity) to predict dropout risk 4 weeks before finals.	Deep learning, LSTM	Dropout forecast timeline
D	Given a student's risk flag and behaviour pattern, an LLM generates a personalised study plan and message from the coordinator. Deploy as React portal with coordinator and student views.	LLM, full-stack deployment	Study plan generator, student portal

Project 21 Automated Essay Scoring and Writing Feedback Tool

Domain: Education — *Dataset:* kaggle: asap-aes

Problem statement: Build a web application for students and teachers where a student submits an essay and receives an instant predicted score, writing style cluster comparison against high-scoring essays, and detailed feedback on content, structure, and vocabulary from an LLM writing coach.

Dataset: ASAP Automated Essay Scoring, Kaggle ([asap-aes competition](#)), 12,977 student essays with human scores.

Student	Sub-problem	Technique	App component
A	Engineer features (sentence length, vocabulary richness, grammar error count). Train Ridge Regression and XGBoost to predict essay score. Metric: Quadratic Weighted Kappa (QWK).	Supervised regression	Score predictor API
B	Cluster essays by writing style profile (vocabulary diversity, sentence complexity). Show which style clusters correlate with high scores.	Unsupervised style clustering	Essay style profiler
C	Fine-tune RoBERTa for essay score prediction using mean token embedding pooling. Report QWK vs feature-engineering baseline.	Transformer regression	BERT scoring service
D	Given an essay and its score, an LLM generates specific feedback on content, structure, and vocabulary with improvement suggestions. Deploy as React web app with essay text editor and score display.	LLM, full-stack deployment	Writing coach interface

Cybersecurity

Project 22 Network Intrusion Detection System

Domain: Cybersecurity — *Dataset:* kaggle: cicdataset/cicids2017

Problem statement: Build a security operations web dashboard that processes network traffic logs, classifies intrusion types in real time, detects anomalous traffic without labels, and generates structured incident reports for security analysts.

Dataset: CICIDS 2017 Dataset, Kaggle (cicdataset/cicids2017), 2.8 million network flows, 14 attack types.

Student	Sub-problem	Technique	App component
A	Train Random Forest and XGBoost on 80 network features to classify 14 attack types. Handle class imbalance with SMOTE.	Supervised multiclass classification	Attack type alert feed
B	Apply Isolation Forest and DBSCAN for unsupervised anomaly detection. Evaluate recall against labelled ground truth.	Unsupervised anomaly detection	Anomaly score dashboard
C	Build a 1D CNN on packet feature sequences for intrusion detection. Compare with tree-based baseline.	Deep learning, 1D CNN	CNN classification service
D	Given detected attack type and affected hosts, an LLM generates a structured SOC incident report with remediation steps. Deploy as FastAPI + React SOC dashboard with live traffic simulation.	LLM, full-stack deployment	Incident report generator, alert queue

Project 23 Phishing URL Detection Browser Extension

Domain: Cybersecurity — *Dataset:* `kaggle: eswarchandt/phishing-website-detector`

Problem statement: Build a browser extension and supporting web API that analyses any URL in real time, classifies it as phishing or legitimate, explains which URL features triggered the alert, and provides an educational explanation of the attack technique. Target users are non-technical employees in organisations.

Dataset: Phishing Websites Dataset (UCI), Kaggle (`eswarchandt/phishing-website-detector`), 11,000 URLs, 30 features.

Student	Sub-problem	Technique	App component
A	Train Random Forest, XGBoost, and SVM on 30 URL features. Target: <0.5% false positive rate (blocking legitimate sites is unacceptable).	Supervised binary classification	URL classification API
B	Cluster phishing URLs by structural attack pattern (domain spoofing, IP-based, subdomain attack). Profile each attack type.	Unsupervised attack pattern mining	Attack pattern explorer
C	Train a character-level CNN/LSTM on raw URL strings without manual features. Show that end-to-end learning is competitive with feature engineering.	Deep learning, character-level model	Character-level classifier
D	Given a detected phishing URL, an LLM generates a plain-language explanation of the attack method. Deliver as a Chrome Extension (Manifest V3) calling a FastAPI endpoint.	LLM, browser extension deployment	Chrome extension popup

Reinforcement Learning

Project 24 Autonomous Game-Playing Agent with Training Dashboard

Domain: RL — *Dataset:* Gymnasium: CartPole-v1 and LunarLander-v2

Problem statement: Build a web-based RL training and visualisation platform that trains agents on increasingly complex control tasks, displays live reward curves and policy visualisations, and generates natural-language training diagnoses when the reward curve plateaus or diverges.

Dataset/Environment: OpenAI Gymnasium — CartPole-v1, LunarLander-v2 (built-in simulators, no dataset download required).

Student	Sub-problem	Technique	App component
A	Implement tabular Q-Learning from scratch on a 5×5 GridWorld. Show Q-table convergence and policy visualisation.	RL, tabular Q-Learning	Policy grid visualiser
B	Train DQN on CartPole-v1 using Stable-Baselines3. Implement experience replay and target network. Plot reward to 500 mean (solved).	RL, DQN	CartPole reward curve
C	Train PPO on LunarLander-v2. Compare PPO vs A2C. Render trained agent behaviour as video. Show policy entropy decay.	RL, PPO, actor-critic	LunarLander live render
D	After each training run, send reward curve statistics to Gemini API for diagnosis and hyperparameter suggestions. Deploy as React dashboard with WebSocket live reward streaming.	LLM, full-stack deployment	Live reward stream, RL coach panel

Project 25 RL-Based Algorithmic Trading Agent

Domain: RL + Finance — *Dataset:* kaggle: rohanrao/nifty50-stock-market-data

Problem statement: Build a web-based paper trading platform that trains a reinforcement learning agent on historical NIFTY 50 data, compares its Sharpe Ratio and drawdown against buy-and-hold and supervised baselines, and integrates a news sentiment pipeline as an additional state variable.

Dataset: NIFTY 50 Stock Market Data, Kaggle (rohanrao/nifty50-stock-market-data); live prices via `yfinance`.

Student	Sub-problem	Technique	App component
A	Build a supervised ML baseline: XGBoost on technical indicators predicting next-day direction. Use as performance upper-bound benchmark.	Supervised, trading baseline	Baseline performance panel
B	Design the Gym-compatible RL environment: state = 30-day OHLCV + position. Action = Buy/Hold/Sell. Reward = daily return minus transaction cost.	RL, environment design	Environment spec documentation
C	Train a DQN agent on the custom trading environment. Evaluate Sharpe Ratio, Max Drawdown, Annualised Return vs Buy-and-Hold.	RL, DQN	Agent performance dashboard
D	Integrate a financial news sentiment pipeline (Gemini API) as an additional state feature. Show whether sentiment-augmented RL outperforms price-only RL on 2023 test data. Deploy as Streamlit paper trading platform.	LLM + RL, full-stack	Paper trading interface, P&L chart

Project 26 Adaptive Traffic Signal Control using RL

Domain: RL + Smart City — *Dataset:* CityFlow open-source simulator

Problem statement: Build a web-based traffic management simulation platform that trains a DQN agent to control a four-way intersection, compares it against a fixed-cycle baseline, and provides a city engineer interface to configure signal plans and view performance metrics.

Dataset/Environment: CityFlow traffic simulation (github.com/cityflow-project); supplementary real traffic data from public city open data portals.

Student	Sub-problem	Technique	App component
A	Analyse historical intersection flow data. Build a regression model predicting average queue length from time-of-day and weather.	Supervised regression	Queue length forecast widget
B	Cluster intersection types by traffic flow pattern using K-Means on hourly counts.	Unsupervised clustering	Intersection profile cards
C	Train a DQN agent in CityFlow: state = per-lane queue lengths; action = signal phase; reward = negative total wait time. Show $\geq 20\%$ wait time reduction vs fixed cycle.	RL, DQN	Live intersection simulation
D	Given hourly congestion data, an LLM generates a traffic bulletin for commuters and timing recommendations for engineers. Deploy as React dashboard with an animated intersection SVG.	LLM, full-stack deployment	Traffic bulletin panel, intersection SVG

NLP and Language

Project 27 AI-Powered Resume Screening and Job Matching Platform

Domain: NLP — *Dataset:* kaggle: gauravduttakiit/resume-dataset + arshkon/linkedin-job-postings

Problem statement: Build a web application for HR teams where recruiters upload a job description and receive a ranked shortlist of matching candidates from a resume pool. Candidates also receive a personalised cover letter draft and gap analysis explaining why they did or did not match.

Dataset: Resume Dataset, Kaggle (gauravduttakiit/resume-dataset), 2,484 resumes across 25 job categories; LinkedIn Job Postings (arshkon/linkedin-job-postings).

Student	Sub-problem	Technique	App component
A	Classify resumes into job category using TF-IDF + SVM and Random Forest. Report top-3 accuracy.	Supervised document classification	Category tagger API
B	Apply K-Means clustering on TF-IDF resume vectors to discover skill-based talent pools. Profile each cluster.	Unsupervised talent clustering	Talent pool explorer
C	Fine-tune Sentence-BERT to create resume and job description embeddings. Rank candidates by cosine similarity. Evaluate NDCG@10.	Deep learning, semantic similarity	Ranked candidate shortlist
D	Given a matched candidate's resume and job description gaps, an LLM generates a tailored cover letter and skill development plan. Deploy as React web app with file upload and candidate table.	LLM, full-stack deployment	Cover letter generator, candidate table

Project 28 Legal Document Summarisation and Q&A System

Domain: NLP — *Dataset:* `kaggle: anadern/indian-court-cases`

Problem statement: Build a web application for law students and paralegals that accepts a legal document upload, automatically classifies the case type, summarises the document, discovers key legal themes, and answers natural-language questions about the document using retrieval-augmented generation.

Dataset: Indian Court Cases, Kaggle (`anadern/indian-court-cases`); EUR-Lex legislative text corpus (`eur-lex.europa.eu`).

Student	Sub-problem	Technique	App component
A	Classify legal documents by case type (criminal, civil, constitutional, taxation) using TF-IDF + Logistic Regression and XGBoost.	Supervised document classification	Case type tagger
B	Apply LDA to discover key legal themes across 500+ judgements. Visualise topic evolution by decade.	Unsupervised topic modelling	Theme explorer, decade timeline
C	Fine-tune LED (Longformer Encoder-Decoder) for legal summarisation. Evaluate ROUGE-1, ROUGE-2, ROUGE-L vs extractive baseline.	Deep learning, abstractive summarisation	Document summary panel
D	Build a legal Q&A assistant using RAG: chunk documents, embed with Sentence-BERT, store in FAISS, answer queries via Gemini API citing specific clauses. Deploy as React web app with PDF upload.	LLM, RAG, full-stack deployment	Q&A chat, clause citation panel

Project 29 Fake News Detection and Fact-Checking Assistant

Domain: NLP — *Dataset:* kaggle: clmentbisaillon/fake-and-real-news-dataset

Problem statement: Build a browser-accessible web application that allows users to paste a news article or headline and receive an authenticity assessment with a confidence score, topic analysis showing whether the article falls into a known disinformation topic cluster, and an LLM-powered fact-check with source pointers.

Dataset: ISOT Fake and Real News Dataset, Kaggle (clmentbisaillon/fake-and-real-news-dataset), 44,898 articles; LIAR Dataset (kaggle: mrisdal/fake-news).

Student	Sub-problem	Technique	App component
A	TF-IDF + Logistic Regression, Naive Bayes, SVM for fake/real binary classification. Analyse top predictive words per class.	Supervised NLP classification	Authenticity score API
B	LDA topic modelling on fake vs real news separately. Identify topics disproportionately represented in fake news.	Unsupervised topic analysis	Topic divergence chart
C	Fine-tune RoBERTa for fake news detection. Implement attention visualisation to show which phrases trigger the fake label.	Transformer fine-tuning	Attention highlight overlay
D	Build an LLM fact-checker: given a snippet, Gemini API identifies inconsistencies and provides a credibility assessment with reasoning. Deploy as React web app with a browser bookmarklet.	LLM, full-stack deployment	Fact-check panel, confidence meter

Social and Creative Applications

Project 30 Social Media Influencer Authenticity Detector

Domain: Social — *Dataset:* `kaggle: free4ever21/instagram-fake-spammer-genuine-accounts`

Problem statement: Build a web application for brand marketing teams that accepts an influencer's handle or profile statistics, classifies the account as genuine or fake, identifies which bot behavioural pattern it matches, and generates a brand safety risk report with a recommendation on whether to proceed with a collaboration.

Dataset: Instagram Fake/Spammer Detection, Kaggle (`free4ever21/instagram-fake-spammer-genuine-accounts`); Twitter Bot Accounts (`kaggle: davidmartngutierrez/twitter-bots-accounts`).

Student	Sub-problem	Technique	App component
A	Engineer engagement features (follower ratio, post frequency, comment/like ratio). Train XGBoost for genuine vs fake classification.	Supervised binary classification	Authenticity score API
B	Cluster accounts by behavioural pattern (low-engagement genuine, high-engagement genuine, bot farm, purchased followers) using K-Means.	Unsupervised behaviour clustering	Behaviour cluster chart
C	Build a Graph Neural Network (PyTorch Geometric) using follower/following relationships as edges to classify account authenticity.	Deep learning, GNN	Network graph visualiser
D	Given authenticity score and cluster profile, an LLM generates a brand safety risk report with red flags and collaboration recommendation. Deploy as React web app with account URL input.	LLM, full-stack deployment	Risk report generator

Project 31 Smart Waste Classification for Recycling

Domain: Environment — *Dataset:* kaggle: asdasdasdasdas/garbage-classification

Problem statement: Build a mobile application for municipal waste management where a citizen photographs a waste item, the app classifies it into the correct bin category (recyclable, organic, hazardous, landfill), highlights the item in the image, and provides disposal instructions and the nearest recycling facility location.

Dataset: Garbage Classification Dataset, Kaggle (asdasdasdasdas/garbage-classification), 2,527 images across 6 waste categories.

Student	Sub-problem	Technique	App component
A	Extract colour histogram and texture features. Train SVM and Random Forest for 6-class waste classification. Non-DL baseline.	Supervised image classification	Baseline accuracy report
B	Apply K-Means on image feature embeddings to cluster waste items. Evaluate whether natural clusters map to recyclability categories.	Unsupervised visual clustering	Cluster visualiser
C	Fine-tune EfficientNet-B0 for waste classification. Convert to TFLite (<200ms on Android). Report accuracy vs model size.	Transfer learning, edge deployment	On-device classifier
D	Given waste category, an LLM generates disposal instructions and environmental impact facts. Deploy as Flutter mobile app with Google Maps integration for facility location.	LLM, mobile deployment	Waste advisor chatbot, facility map

Project 32 Music Discovery and Mood-Based Recommendation App

Domain: Creative — *Dataset:* kaggle: andradaolteanu/gtzan-dataset

Problem statement: Build a web music discovery application where users describe their mood or activity in natural language, the system maps the description to audio feature constraints, retrieves matching tracks from a genre-clustered library, and generates a personalised playlist with an LLM-written playlist description.

Dataset: GTZAN Genre Collection, Kaggle (andradaolteanu/gtzan-dataset-music-genre-classification), 1,000 tracks across 10 genres with precomputed MFCC and spectral features.

Student	Sub-problem	Technique	App component
A	Use precomputed MFCC and spectral features. Train SVM and Random Forest for 10-genre classification. Per-genre confusion matrix.	Supervised audio classification	Genre classifier API
B	Apply PCA and K-Means on audio features. Evaluate whether discovered clusters match genre labels using ARI score.	Unsupervised audio clustering	Genre cluster explorer
C	Build a 2D CNN on Mel spectrogram images (audio-to-image conversion). Compare accuracy vs feature-engineering baseline.	Deep learning, CNN on spectrograms	Spectrogram CNN service
D	User types a mood description; Gemini API maps it to genre and tempo constraints; recommender returns tracks; LLM writes a playlist description. Deploy as React music app with audio preview.	LLM, full-stack deployment	Mood-to-playlist interface

Project 33 Devanagari Handwriting Recognition and Digitisation Tool

Domain: Creative — *Dataset:* kaggle: rishianand/devanagari-character-set

Problem statement: Build a web application that digitises handwritten Devanagari text from uploaded images, corrects OCR errors using language context, and exports clean digital text. Target users include government clerks digitising historical records and students submitting handwritten assignments.

Dataset: Devanagari Handwritten Character Dataset, Kaggle (rishianand/devanagari-character-set), 92,000 character images across 46 classes.

Student	Sub-problem	Technique	App component
A	Extract HOG and pixel intensity features. Train SVM and Random Forest on Devanagari characters. Establish non-DL accuracy baseline.	Supervised image classification	Baseline character recogniser
B	Apply t-SNE to visualise character embeddings. Identify visually similar character groups the model may confuse.	Unsupervised embedding visualisation	Character confusion map
C	Fine-tune ResNet-18 on character images. Implement CTC loss layer for full word recognition (sequence output from an image strip).	Deep learning, CNN + CTC	Word recognition service
D	Pass raw OCR text through Gemini API with Hindi language context to correct recognition errors. Deploy as React web app with image upload and side-by-side original/corrected text.	LLM OCR correction, full-stack	Image upload, text correction panel

Project 34 Sports Performance Analytics Platform

Domain: Sports — *Dataset:* `kaggle: patrickb1912/ipl-complete-dataset-20082020`

Problem statement: Build a web application for cricket team management that predicts match outcomes and individual player performance, segments players by playing style, forecasts next-ball outcomes, and generates pre-match tactical briefings for coaching staff.

Dataset: IPL Complete Dataset, Kaggle (`patrickb1912/ipl-complete-dataset-20082020`); NBA Advanced Player Statistics (`kaggle: justinas/nba-players-data`).

Student	Sub-problem	Technique	App component
A	Predict match outcome and individual batting/bowling performance using XGBoost and Random Forest on historical match and player features.	Supervised classification + regression	Match prediction widget
B	Cluster players by batting style (aggressive opener, anchor, death-over finisher) using K-Means on career statistics. Build player similarity finder.	Unsupervised player profiling	Player similarity explorer
C	Train an LSTM on ball-by-ball sequence data to predict next-ball outcome (dot/1/2/4/6/wicket) as sequence classification.	Deep learning, LSTM	Ball-by-ball predictor
D	Given a match situation (scorecard, pitch report, opposition analysis), an LLM generates a pre-match tactical briefing and commentary. Deploy as React sports analytics dashboard.	LLM, full-stack deployment	Tactical briefing panel, scorecard

Project 35 Real Estate Price Prediction and Valuation Platform

Domain: Smart City — *Dataset:* `kaggle: house-prices-advanced-regression-techniques`

Problem statement: Build a web application for real estate agents and home buyers that predicts property valuations, identifies neighbourhood clusters with similar value profiles, explains the key drivers of each property's price, and generates a natural-language property valuation report.

Dataset: House Prices: Advanced Regression Techniques, Kaggle (`house-prices-advanced-regression-techniques`), 79 features across 1,460 training properties.

Student	Sub-problem	Technique	App component
A	Handle 79 mixed features (36 numeric, 43 categorical). Train Ridge, Lasso, XGBoost. Log-transform price. Target: competitive RMSLE on Kaggle leaderboard.	Supervised regression	Price prediction API
B	Cluster properties by location and feature profile using K-Means. Build a neighbourhood value map.	Unsupervised segmentation	Neighbourhood cluster map
C	Train a deep tabular model with entity embeddings for categorical features. Implement SHAP for individual price explanation.	Deep learning tabular, XAI	SHAP feature chart
D	Given property features and SHAP values, an LLM generates a natural-language valuation report and listing description. Deploy as React web app with a property entry form.	LLM, full-stack deployment	Valuation report generator, property form

Project 36 Flood Risk Prediction and Early Warning System

Domain: Environment — *Dataset:* kaggle: nataliaivanova7/global-flood-database

Problem statement: Build a web-based disaster management platform for district authorities that predicts flood risk by district, visualises historical flood event clusters, segments areas by risk severity, and auto-generates multi-channel early warning alerts (SMS text, email, broadcast format) when predicted risk exceeds a threshold.

Dataset: Global Flood Database, Kaggle (nataliaivanova7/global-flood-database); India flood data from `data.gov.in`.

Student	Sub-problem	Technique	App component
A	Predict flood risk (binary) using XGBoost and Random Forest on rainfall, river proximity, elevation, and historical frequency features.	Supervised binary classification	District risk predictor
B	Cluster historical flood events by severity, spatial extent, and season using DBSCAN. Profile each flood typology.	Unsupervised event clustering	Flood event cluster map
C	Build a U-Net CNN for semantic segmentation of flood-prone zones from rainfall heatmap tiles.	Deep learning, image segmentation	Flood zone segmentation overlay
D	Given flood risk predictions per district, an LLM generates SMS-format evacuation alerts and infrastructure action plans. Deploy as Leaflet.js web dashboard with alert simulation.	LLM, geospatial, full-stack	Alert generator, district risk map

Project 37 Personalised Learning Path Recommender for Online Education

Domain: Education — *Dataset:* `kaggle: DataTalks.Club/course-recommendations`

Problem statement: Build a web application for an EdTech platform that profiles a learner's skills and goals from a short intake form, recommends a personalised course sequence using collaborative and content-based filtering, predicts completion likelihood for each recommended path, and generates a week-by-week study plan.

Dataset: Coursera Course Dataset (`kaggle: khusheekapoor/coursera-courses-dataset`); MOOC user interaction logs (`kaggle: nhalai/coursera-learners-data`).

Student	Sub-problem	Technique	App component
A	Predict course completion likelihood (binary) from learner engagement features using Logistic Regression and XGBoost.	Supervised binary classification	Completion likelihood badge
B	Cluster courses by topic and difficulty using TF-IDF + K-Means on course descriptions. Build a course similarity network.	Unsupervised content clustering	Course cluster explorer
C	Train Neural Collaborative Filtering (NCF) in PyTorch on learner-course interaction matrix for personalised ranking. Compare with SVD.	Deep learning, neural recommender	Personalised course list
D	Given learner profile and recommended courses, an LLM generates a week-by-week study plan with time estimates and milestone checkpoints. Deploy as React EdTech app with progress tracker.	LLM, full-stack deployment	Study plan generator, progress tracker

Project 38 Multi-Modal Product Search and Visual Discovery Engine

Domain: Retail — *Dataset:* `kaggle: agrigorev/clothing-dataset-full`

Problem statement: Build a fashion e-commerce web application that allows users to either upload an image of a clothing item or describe what they want in text, retrieves visually and semantically similar products, and generates a personalised style note explaining why each result matches the user's query.

Dataset: Clothing Dataset, Kaggle (`agrigorev/clothing-dataset-full`), 5,000 clothing images with category labels; Fashion Product Images (`kaggle: paramaggarwal/fashion-product-images-dataset`).

Student	Sub-problem	Technique	App component
A	Train a classifier (SVM + HOG features, then Random Forest) for clothing category prediction. Build a category filter component.	Supervised image classification	Category filter API
B	Extract CNN embeddings from a pre-trained model. Apply K-Means on embeddings to cluster visually similar styles. Build a visual cluster browser.	Unsupervised visual clustering	Style cluster browser
C	Fine-tune CLIP (Contrastive Language-Image Pre-training) for joint text-image embedding space. Enable text-to-image and image-to-image retrieval.	Deep learning, multi-modal retrieval	Dual-search retrieval engine
D	Given the user's text query and retrieved results, an LLM generates a personalised style note explaining each match. Deploy as React storefront with image upload and text search.	LLM, full-stack deployment	Style note generator, product grid

Summary of All Projects

#	Project Title	Domain	Dataset (Kaggle unless noted)
1	Cardiovascular Disease Risk Screening	Healthcare	sulianova/cardiovascular-disease-dataset
2	Chest X-Ray Pneumonia Detection	Healthcare	paultimothymooney/chest-xray-pneumonia
3	Mental Health Risk Detection from Text	Healthcare	neelghoshal/reddit-mental-health-data
4	Drug Side Effect and Review Intelligence	Healthcare	jessicali9530/kuc-hackathon-winter-2018
5	Real-Time Credit Card Fraud Detection	Finance	mlg-ulb/creditcardfraud
6	Stock Market Signal Intelligence Platform	Finance	camnugent/sandp500 + yfinance
7	Loan Default Prediction and Credit Scorecard	Finance	home-credit-default-risk
8	Insurance Risk Assessment and Advisory	Finance	mirichoi0218/insurance
9	Predictive Maintenance for Industrial Equipment	Manufacturing	stephanmatzka/predictive-maintenance-dataset-ai4i-2020
10	Visual Quality Control for Defects	Manufacturing	ipythonx/mvtec-ad
11	Energy Consumption Forecasting	Manufacturing	uciml/electric-power-consumption-data-set
12	Supply Chain Demand Forecasting	Manufacturing	rossmann-store-sales
13	Crop Disease Detection Mobile App	Agriculture	emmarex/plantdisease
14	Air Quality Forecasting and Alert System	Environment	rohanrao/air-quality-data-in-india
15	Ride-Hailing Demand Forecasting	Mobility	nyc-taxi-trip-duration
16	Road Accident Severity and Hotspot Analysis	Mobility	tsiaras/uk-car-accidents-2005-to-2007
17	Customer Churn Prediction and Retention	Retail	blastchar/telco-customer-churn
18	Intelligent Product Recommendation System	Retail	snap/amazon-fine-food-reviews
19	Customer Review Sentiment Intelligence	Retail	kazanova/sentiment140

#	Project Title	Domain	Dataset
20	Student Performance and Early Intervention	Education	larsen0966/student-performance-data-set
21	Automated Essay Scoring and Feedback	Education	asap-aes (Kaggle competition)
22	Network Intrusion Detection System	Cybersecurity	cicdataset/cicids2017
23	Phishing URL Detection Browser Extension	Cybersecurity	eswarchandt/phishing-website-detector
24	Autonomous Game-Playing Agent	RL	Gymnasium (CartPole-v1, LunarLander-v2)
25	RL-Based Algorithmic Trading Agent	RL + Finance	rohanrao/nifty50-stock-market-data
26	Adaptive Traffic Signal Control (RL)	RL + Smart City	CityFlow open-source simulator
27	AI-Powered Resume Screening and Matching	NLP	gauravduttakiit/resume-dataset
28	Legal Document Summarisation and Q&A	NLP	anaderm/indian-court-cases
29	Fake News Detection and Fact-Checking	NLP	clmentbisailon/fake-and-real-news-dataset
30	Social Media Influencer Authenticity	Social	free4ever21/instagram-fake-spammer-genuine-accounts
31	Smart Waste Classification for Recycling	Environment	asdasdasdasdas/garbage-classification
32	Music Discovery and Mood Recommendation	Creative	andradaolteanu/gtzan-dataset-music-genre-classification
33	Devanagari Handwriting Recognition	Creative	rishianand/devanagari-character-set
34	Sports Performance Analytics Platform	Sports	patrickb1912/ipl-complete-dataset-20082020
35	Real Estate Price Prediction Platform	Smart City	house-prices-advanced-regression-techniques
36	Flood Risk Prediction and Early Warning	Environment	nataliaivanova7/global-flood-database
37	Personalised Learning Path Recommender	Education	khusheekapoor/coursera-courses-dataset
38	Multi-Modal Product Search and Discovery	Retail	agrigorev/clothing-dataset-full

All datasets are publicly available at no cost. Kaggle datasets require a free Kaggle account.

LLM API: Google AI Studio (Gemini 1.5 Flash) provides a free tier sufficient for all Student D sub-problems.

Deployment: Render.com, Railway.app, and HuggingFace Spaces all offer free hosting tiers for student projects.